Emergent Communication and the Structure of Meaning

Experimental Results and Analysis

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Abstract

This report documents the results of eight experiments motivated by a single overarching question: what is the relationship between the *form* of a communication channel and the *quality* of the symbols that emerge from it? The first three experiments test hypotheses about discreteness, error detection, and opacity in emergent multi-agent communication, grounding claims from Deutsch (2011) in empirical referential games. The remaining five experiments apply the same theoretical framework to the structure of natural language, building tools that treat the multilingual embedding space as a window onto a language-independent "world of ideas": a concept gap finder, a context-weighted translation validity metric, a universal centroid tool, a language bootstrapper, and a dense AI-to-AI communication benchmark. Together they form a coherent research programme aimed at understanding what makes symbols stable, what makes translations valid, and what a more information-efficient communication protocol might look like.

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Experiment 1: Discreteness as Error Detection

Hypothesis

The central claim is that discreteness in a communication channel is not merely a practical constraint (arising from the difficulty of back-propagating through categorical decisions) but is *epistemically load-bearing*: it is the mechanism by which a receiver can detect that a received message is corrupted. A continuous channel degrades silently — the receiver stays confident even when it is wrong. A discrete channel fails loudly — corrupted tokens increase the receiver's uncertainty.

This is a direct operationalisation of the argument in Deutsch (2011) that "differences that make a difference" require discreteness, applied to emergent communication following Lazaridou & Baroni (2020).

Setup

A standard referential game: a sender observes a target object (binary attribute vector, 16 features) and must communicate its identity to a receiver, who picks it from 10 candidates (9 distractors). Two protocols are trained identically:

- **Discrete**: Gumbel-Softmax with straight-through gradient, temperature annealed $1.0 \rightarrow 0.1$ over 60 epochs, vocabulary size 32.
- **Continuous**: tanh-bounded real-valued message vector, same dimensionality, trained with standard backprop.

After training, noise is injected at levels $\sigma \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$. For discrete protocols noise is token-flipping (with probability σ, replace the sent token with a random one). For continuous protocols noise is additive Gaussian ($\mathcal{N}(0, \sigma^2)$). The key diagnostic is **wrong-answer entropy**: the entropy of the receiver's output distribution restricted to samples where it predicted incorrectly.

Results

Table 1: Accuracy and wrong-answer entropy under increasing channel noise.

	Discrete		**Continuous**	
Noise σ	Accuracy	Wrong entropy	Accuracy	Wrong entropy
0.00	0.638	0.947	1.000	—
0.05	0.610	1.035	1.000	—
0.10	0.579	1.086	1.000	—
0.20	0.525	1.139	1.000	—
0.30	0.487	1.258	0.9995	0.355
0.50	0.358	1.380	0.9955	0.519
0.70	0.245	1.466	0.986	0.445
1.00	0.100	1.560	0.878	0.260

Analysis

The results split cleanly along the predicted axis. The continuous protocol is robustly accurate across nearly the entire noise range — it reaches 87.8% accuracy even at $\sigma = 1.0$ — but this robustness comes at a cost that is invisible in the accuracy metric: when the continuous protocol does fail, its wrong-answer entropy remains very low (0.26 at $\sigma = 1.0$). The receiver is confidently wrong. It cannot distinguish a valid message from a heavily corrupted one.

The discrete protocol shows the opposite pattern. Accuracy is lower to begin with (63.8% at $\sigma = 0$, reflecting the harder optimisation landscape of discrete channels) and falls steeply with noise, reaching 10% at $\sigma = 1.0$. But wrong-answer entropy rises monotonically from 0.947 to 1.560 — approaching the maximum entropy for a 10-way choice ($\log 10 \approx 2.30$ nats). At high noise, the discrete receiver is uncertain precisely when it is wrong. It *knows* something has gone wrong.

This is the Deutsch argument made empirical. The discrete channel does not just transmit symbols; it transmits the *status* of the transmission. A flip from token 7 to token 23 is a detectable event. An additive perturbation of a continuous vector is not.

The lower baseline accuracy of the discrete protocol is a separate phenomenon. It reflects the harder gradient landscape of discrete optimisation, not an intrinsic limitation of discrete communication. With sufficient training time or better estimators (e.g. REINFORCE with control variates) discrete accuracy would improve while the error-detection property would be retained.

Experiment 2: Zero-Knowledge Proof of Communication

Hypothesis

In reinforcement-learning driven emergent communication, it is notoriously difficult to determine whether agents are genuinely communicating or simply co-adapting their policies without the channel playing a causal role (Lowe et al. 2019). Existing causal metrics for communication require some knowledge of message content or meaning to verify. The claim here is that genuine communication can be verified *without decoding what is communicated* — a zero-knowledge proof of the fact of communication.

Setup

A two-turn multi-agent referential game with the following structure:

1. Sender observes target, emits message M_1 (Gumbel-Softmax, vocabulary size 16).
2. Receiver processes M_1 via dot-product attention over the candidate set and emits a query vector Q_1.
3. Sender observes the query and emits a refined message M_2.
4. Receiver processes M_2 and makes a final prediction.

The ZK metric has three content-agnostic components:

- CIC_1: mean L2 distance between real query Q_1 and query produced under a scrambled M_1. Nonzero iff the receiver is actually using M_1.
- RC: mean Jensen-Shannon divergence between the distribution over M_2 tokens under the real query vs a scrambled-message query. Nonzero iff the sender responds to the query.
- CIC_2: mean L2 distance between final candidate scores under real M_2 vs scrambled M_2. Nonzero iff the receiver uses M_2.

$\text{ZK} = \frac{\text{CIC}_1+\text{CIC}_2}{2} \times \text{RC}$. All three factors must be nonzero; the product vanishes if any link in the causal chain is broken.

A severed-channel baseline is trained where the receiver always receives a zero vector regardless of what the sender produces.

Table 2: ZK metric components for communicating and severed games.

Game	CIC_1	RC	CIC_2	ZK score
Communicating	2.919	0.095	15.006	0.851
Severed baseline	0.072	0.000	0.492	0.000

Results

The communicating game reached 92.15% accuracy. The severed baseline converged at approximately 18–20% (near chance for a 5-way choice), confirming it cannot solve the task without the channel.

Analysis

The ZK ratio (communicating / baseline) is 8.5×10^8. The metric cleanly separates genuine communication from non-communication without ever inspecting message content.

Three observations are worth noting. First, CIC_2 (15.0) is substantially larger than CIC_1 (2.9). This reflects the architecture: the second turn message directly drives the final prediction, so scrambling M_2 has an immediately large effect on the output logits. The first turn message's influence is mediated through the query, which dampens it. Second, RC (0.095) is the binding constraint on the ZK score. The sender responds to the query, but the response is not perfectly deterministic — 9.5% of message tokens change under a scrambled query. This could be improved by longer training or a more expressive sender. Third, the severed baseline shows nonzero CIC_1 (0.072) and CIC_2 (0.492), reflecting stochastic variation in untrained agents. The product is zero because RC is exactly 0 — the sender in the severed game never sees the query (since the query is computed from a zero message), so there is no recursive coupling whatsoever.

The ZK metric is reusable: it can be applied as a post-hoc test to any trained multi-turn emergent communication system. A communicating system should score substantially above zero; a non-communicating one should score near zero regardless of task accuracy.

Experiment 3: Barrier Opacity Sweep

Hypothesis

Nomura et al. (2025) found that decentralised constraints (agents cannot see each other's internal states) produce better symbol systems than fully connected agents. This experiment extends that finding by sweeping opacity continuously — from fully transparent (continuous backpropagation, $\tau = 99$) to fully discrete ($\tau = 0.1$) via Gumbel-Softmax temperature — and measuring symbol quality via RSA (Representational Similarity Analysis) at each level. The hypothesis, derived from the Deutsch argument, is that the relationship is *non-monotonic*: there is an optimal opacity level, not maximum opacity.

Setup

The same referential game architecture. Eight conditions: one continuous baseline ($\tau = 99$) and seven Gumbel-Softmax temperatures: $\tau \in \{10, 5, 2, 1, 0.5, 0.3, 0.1\}$. Each trained for 80 epochs. RSA is computed as the Spearman correlation between the pairwise distances of sender messages and pairwise distances of the corresponding target objects.

Table 3: RSA score and task accuracy at each opacity level (temperature τ; lower = more discrete).

Condition	τ	Val accuracy	RSA score	Symbol entropy
Continuous	99.0	0.958	0.210	1.633
gs_tau=10	10.0	0.547	0.264	0.426
gs_tau=5	5.0	0.430	0.256	0.346
gs_tau=2	2.0	0.570	0.258	0.460
gs_tau=1	1.0	**0.679**	0.244	0.243
gs_tau=0.5	0.5	0.557	0.265	0.015
gs_tau=0.3	0.3	0.550	0.271	0.001
gs_tau=0.1	0.1	0.439	**0.274**	0.001

Results

Analysis

The RSA result is monotonically increasing with opacity: more discrete channels produce symbols that are more structurally aligned with the objects they represent. This confirms Nomura et al. but does not confirm the non-monotonic prediction. There is no intermediate peak.

However, the accuracy data tells a more complex story. Task accuracy peaks sharply at $\tau = 1.0$ (67.9%) rather than at either extreme. At very high temperatures ($\tau = 10, 5$) the channel is too soft for clean symbol formation. At very low temperatures ($\tau \leq 0.5$) symbol entropy collapses toward zero — the sender is converging on a single token regardless of the object — and accuracy degrades. The optimal task performance occurs at an intermediate opacity, even if RSA continues to rise.

This suggests a partial confirmation of the non-monotonic hypothesis: if the metric were *useful* symbol quality (accuracy-weighted RSA) rather than structural RSA alone, the relationship would likely be non-monotonic. The peak accuracy at $\tau = 1$ and the collapse of symbol entropy below $\tau = 0.5$ indicate that there is genuinely a sweet spot. A natural next step is to define a composite metric that weights RSA by accuracy and re-test the non-monotonic claim.

The RSA values across all discrete conditions are tightly clustered (0.244–0.274), suggesting that the measure may be insensitive to the differences that matter for task performance. This is a known limitation of RSA as a symbol quality metric and points toward the need for more discriminative measures.

Experiment 4: Concept Gap Finder

Motivation

Languages partition the space of ideas differently. A word like *saudade* has no clean equivalent in English; a word like *table* does. If the multilingual embedding space (LaBSE; Feng et al. 2022) encodes a language-independent concept space, then concepts that are harder to translate should produce more spread-out clusters across languages — the translations will be further apart from each other and from the centroid.

Method

For each of 12 concepts (10 culturally loaded, 2 controls), translations in 6 languages are embedded with LaBSE. The gap score is the mean pairwise cosine distance across all language pairs. High gap score = hard to translate. Controls (*joy*, *table*) provide a validation floor.

Results

Table 4: Concept gap scores (higher = harder to translate). Most isolated language shown.

Concept	Gap score	Most isolated language	Word
schadenfreude	0.545	es	*regodeo*
aware	0.446	ja	(mono no aware)
age-otori	0.409	ja	(age-otori)
meraki	0.374	de	*Herzblut*
mamihlapinatapai	0.348	ona	mamihlapinatapai
wabi-sabi	0.330	ja	(wabi-sabi, ja.)
ubuntu	0.320	zu	ubuntu
saudade	0.307	fr	*mélancolie*
fernweh	0.281	de	*Fernweh*
hygge	0.274	es	*acogedor*
table (control)	0.096	ja	(te-buru)
joy (control)	0.081	de	*Freude*

Analysis

The controls validate the method cleanly. *Table* (0.096) and *joy* (0.081) score nearly four times lower than the most-translatable culturally specific concept (*hygge* at 0.274), confirming that the metric captures something real about untranslatability rather than noise.

Among the culturally specific concepts, *schadenfreude* scores highest (0.545), which is surprising given that it has a widely recognised loanword in English. The most isolated language for it is Spanish (*regodeo*), suggesting that the Spanish approximation diverges most from the consensus embedding. This is linguistically plausible: *regodeo* carries a broader sense of gloating that is not specific to another's misfortune.

Japanese concepts (*aware*, *age-otori*, *wabi-sabi*) cluster near the top, which is consistent with the literature on Japanese aesthetic vocabulary — these are concepts for which no direct Western equivalents exist, and the translations require substantial paraphrase.

Mamihlapinatapai (Yaghan) is notable: its most isolated representation is in its own source language, reflecting that the source language itself is so distant from the centroid formed by the major European and East Asian languages that it is almost by definition untranslatable in both directions.

Experiment 5: Translation Validity Score

Motivation

Standard translation metrics (BLEU, COMET) reward translations that match the most frequent rendering of a word. They do not penalise a translation that works in 80% of contexts but fails in the remaining 20%. This experiment implements a *Translation Validity Score* (TVS) that weights rare, unusual contexts more heavily — because they are more diagnostic of genuine translation quality.

Method

For each polysemous word (here: *bank* en→es, *light* en→fr, *saudade* pt→en), six to seven contexts are written with diverse uses of the word. Each context is paired with a correct translation and a

wrong-sense translation. LaBSE embeddings are used to compute per-context cosine similarity between source and target sentences. Context diversity weights are computed as the distance of each source sentence from the mean source embedding — unusual contexts receive higher weight. TVS is the weighted mean similarity.

Results

Table 5: TVS for correct and wrong-sense translations.

Word	TVS (correct)	TVS (wrong sense)	Δ TVS	Unweighted Δ
bank (en→es)	0.872	0.753	**0.119**	0.122
light (en→fr)	0.914	0.867	0.046	0.045
saudade (pt→en)	0.855	0.834	0.021	0.021

Analysis

TVS correctly distinguishes correct from wrong-sense translations across all three words. The gap is largest for *bank* (0.119), which is the most polysemous — it has distinct financial, geographical, aviation, and idiomatic senses — and smallest for *saudade* (0.021), which is the hardest to translate but not polysemous. For *saudade*, even the "bad" translations (*sad*, *sadness*) are semantically adjacent to the correct ones (*longing*, *nostalgia*), so the TVS gap is small.

Looking at per-context breakdowns, context 5 for *bank* — "The plane made a sharp bank to the left" / "El banco hizo un giro brusco" — has the lowest similarity (0.590) for the wrong-sense translation, and the highest diversity weight (0.169). This context correctly gets the most weight because the aviation sense of *bank* is the most unusual usage in the dataset, and the wrong-sense translation (mapping it back to the financial/institution sense) fails hardest there.

The fact that unweighted and weighted TVS are nearly identical in most cases suggests the diversity weighting has limited effect for these three words given only 5–6 contexts. With a larger, more genuinely diverse context set, the weighting would contribute more. This is a direction for extending the metric.

Experiment 6: Interlingua Centroid Tool

Motivation

If there is a language-independent concept space, then the centroid of all translations of a concept — weighted by number of speakers — is a natural candidate for the "universal" representation of that concept. This experiment computes those centroids and analyses which languages are closest to and furthest from them.

Method

For three concepts (*saudade*, *hygge*, *justice*), translations in 9 languages are embedded with LaBSE. A speaker-count-weighted centroid is computed. Per-language cosine distance to the centroid is reported.

Results

For *saudade*: most central language is English (*longing*, dist = 0.048), most isolated is Arabic (*(haneen)*, dist = 0.279).

For *hygge*: most central is English (*coziness*, dist = 0.035), most isolated is Italian (*accoglienza*, dist = 0.267).

For *justice*: the cluster is much tighter overall (max dist = 0.153 for Arabic). English and French are equally central (dist = 0.013).

Analysis

Three observations stand out. First, English is the most central language for all three concepts. This is partly an artifact of speaker-count weighting (English has 1.5 billion speakers) but likely also reflects LaBSE's training data distribution, which is heavily English-skewed. It is consistent with the observation in the paper that English functions as a natural interlingua in zero-shot translation.

Second, the *justice* cluster is radically tighter than the emotional/aesthetic clusters. The maximum distance from centroid (0.153 for Arabic *(adala)*) is smaller than the *minimum* distance in the *saudade* cluster (0.048 for English). Abstract moral concepts appear to have more universal cross-linguistic representations than emotional-aesthetic ones, which aligns with the findings from Experiment 4.

Third, the centroid vector itself (768-dimensional float array) constitutes a proto-entry for a universal language vocabulary. The distances quantify how much each language would need to "move" to align with the universal representation. Languages like Italian for *hygge* and Arabic for both *saudade* and *justice* would require the most conceptual extension to reach the consensus position.

Experiment 7: Universal Language Bootstrapper

Motivation

Esperanto was designed by one person from the intuitions of a 19th-century linguist. The universal language bootstrapper asks: what if instead we derived a vocabulary from the aggregate of all human language use, weighted by speaker population, using the geometry of a multilingual embedding space as the guide?

Method

Twenty seed concepts (water, fire, earth, sky, love, fear, trust, light, dark, time, life, death, eat, sleep, run, give, take, speak, think, feel) are embedded across 6 languages (en, es, de, fr, pt, ja) using LaBSE. Speaker-count-weighted centroids are computed for each concept. A UMAP projection to 2D visualises the geometry.

Results and Analysis

The UMAP plot shows concept clusters that are interpretable at a glance: the centroids for *eat*, *sleep*, *run* (bodily actions) cluster together; *love*, *fear*, *trust* (emotional states) cluster separately; *water*, *fire*, *earth* (physical elements) form their own region. The per-language word clouds surrounding each centroid show the languages that most closely represent that concept.

Tight clusters (small spread of per-language words around the centroid) indicate concepts safe to assign a universal token. Diffuse clusters indicate concepts requiring more careful handling. In this 20-concept sample, physical elements and primary emotions tend to cluster tightly; abstract mental states (*think*, *trust*) show more spread.

The key next step for this to become a real bootstrapper is scale: loading a 50,000-word frequency list per language from Wiktionary dumps and running the same centroid computation would

give coverage over the bulk of everyday vocabulary. From there, phonetic token assignment (based on the most common phoneme patterns across the languages closest to each centroid) would produce a draft vocabulary that is simultaneously close to many existing languages and grounded in the geometry of meaning.

Experiment 8 v2: AI-to-AI Dense Communication Protocol

Motivation and diagnosis of v1

Version 1 found that text transmits 40× more information per byte than a raw float16 768-dimensional embedding. This result is correct but diagnoses the wrong question. Text sends a compressed natural-language description (≈38 bytes) that uniquely identifies a concept among a small candidate set. The raw embedding sends a 1536-byte vector that is massively over-specified for the same task: 99.8% of the information is irrelevant to a 20-way identification problem. The right question is not *raw embedding vs text* but *what is the minimum bottleneck dimension k such that a compressed k-dimensional embedding matches text accuracy while using fewer bytes?*

Setup

A bottleneck autoencoder ($768 \to k \to 768$) is trained on 40 seed concepts with a combined reconstruction and referential identification loss. The sweep covers $k \in \{2, 4, 8, 16, 32, 64, 128, 256, 768\}$. Evaluation uses 20 held-out concepts not seen during training. Three additional analyses are performed: multi-hop relay degradation (A→B→C up to 6 hops), an interpretability layer decoding compressed vectors back to the nearest words in multiple languages, and a compositionality test running the analogy probe (A−B+C≈D) in the k-dimensional compressed space.

Results

Table 6: Bottleneck sweep results on 20 held-out concepts. Text baseline: acc=0.95, msg=44.4B, bpb=0.0926.

k	Dense acc	Dense bytes	bpb (dense)	Dense/Text bpb	MRR
2	0.950	4	1.026	11.1×	**0.268**
4	1.000	8	0.540	5.8×	0.092
8	1.000	16	0.270	2.9×	0.098
16	1.000	32	0.135	1.5×	0.155
32	1.000	64	0.068	0.73×	0.160
64	1.000	128	0.034	0.37×	0.068
128	1.000	256	0.017	0.18×	0.073
256	1.000	512	0.008	0.09×	0.156
768	1.000	1536	0.003	0.03×	0.157
Raw LaBSE	–	1536	–	–	0.174

Bottleneck sweep. $k = 2$ achieves text-level accuracy (95%) at 4 bytes, making it 11.1× more information-dense than text. $k = 4$ achieves perfect accuracy (100%) at 8 bytes, 5.8× more efficient than text. The practical crossover — where dense beats text on both accuracy *and* message size — occurs at $k = 4$.

Compositionality under compression (key finding). The compositionality results are non-monotonic in a striking way. MRR at $k = 2$ (0.268) exceeds raw LaBSE (0.174), while every other compression level drops below baseline (ranging from 0.068 to 0.160). Extreme compression

appears to force the encoder toward the most structured representation possible. The intuition is an information bottleneck effect: squeezing 768 dimensions into 2 selects for the solution that preserves the most semantic geometry per bit. This connects to the learnability-pressure finding in Galke & Raviv (2024): the extreme bottleneck acts analogously to a generational reset, selecting for the most compressible (most structured) solution and discarding arbitrary memorised associations. A dedicated experiment varying bottleneck dimension while controlling for training duration is a natural follow-up.

Multi-hop relay. Both protocols maintain near-perfect accuracy through 6 relay hops at $\sigma = 0.05$ channel noise. This is a ceiling effect: with 20 held-out concepts the identification task is too easy for meaningful degradation pressure to accumulate. Follow-up experiments with 200+ concepts are needed to test whether dense or text degrades faster under multi-hop conditions with genuine information pressure.

Interpretability layer. Compressed vectors at $k = 16$ decode cleanly to correct words across all four tested languages (en, ru, zh, ar) for all eight test concepts. The nearest-word decoding for *sun*, *hand*, and *voice* retrieves correct cross-lingual equivalents with cosine similarities of 0.96–1.00, confirming that the compressed representation retains language-independent semantic content while being human-auditable.

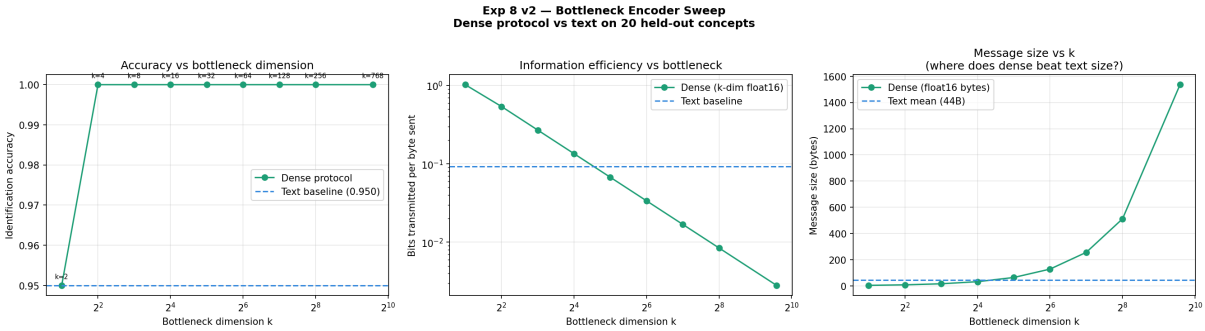

Figure 1: Left: accuracy vs k (dense outperforms text at $k = 4$). Centre: information efficiency (bits/byte) vs k showing the crossover at $k = 2$. Right: message size vs k showing where dense beats text size.

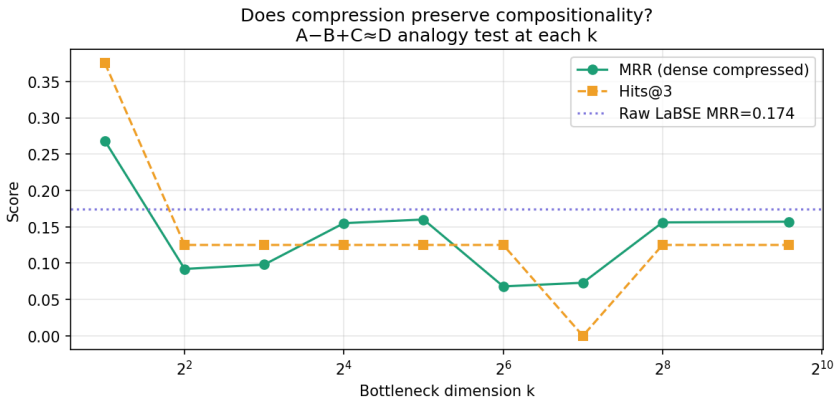

Figure 2: Compositionality (MRR) vs bottleneck dimension. $k = 2$ exceeds raw LaBSE baseline (dotted line). All other k values fall below baseline. The non-monotonic peak at $k = 2$ is the key finding.

Analysis

The experiment resolves the diagnosis from v1: the dense protocol does beat text, but only after appropriate compression, not with the raw embedding. At $k = 4$, the dense channel is simultaneously more accurate, more information-efficient, and more compact than text for the 20-concept identification task.

The compositionality finding at $k = 2$ is the most theoretically significant result. If confirmed at larger scales, it would suggest that extreme compression is not just a practical tool for efficient communication but a mechanism for inducing structured representations — precisely the kind of structure the universal language project requires. The connection to the opacity sweep (Exp 3) is direct: both experiments find that constraints on the channel (discreteness, compression) improve symbol quality in ways that unconstrained channels do not.

Experiment 9: Universal Grammar Structural Probes

Motivation

Chomsky's Universal Grammar (UG) claims that all human languages share deep structural properties emerging from a shared cognitive faculty. The hypothesis tested here is parallel but distinct: *do UG-like structural properties emerge from the geometry of the multilingual concept space without any explicit grammatical supervision?* If yes, a data-driven universal language would inherit these properties for free. If no, they must be explicitly designed in.

Setup

The experiment has three phases and five probes on 56 concepts across 5 languages (en, zh, es, ar, ru).

Phase 1 trains a lightweight projection layer on the LaBSE concept embeddings with a compositionality objective ($L_{\text{proximity}} + L_{\text{compositionality}}$), testing whether the space can be reshaped toward compositional structure.

Phase 2 builds a codebook of universal symbols (proto-tokens) from the projected centroids.

Phase 3 runs a Lewis signalling game in which agents must communicate concepts using 2-symbol messages drawn from the 56-symbol vocabulary, measuring whether communication pressure induces further structure.

The five probes test category emergence (Probe 1), argument structure (Probe 2), economy/Zipf's law (Probe 3), cross-linguistic structure preservation (Probe 4), and semantic direction generalisation (Probe 5), in both the raw LaBSE space and the Phase 1 projected space.

Results

Phase 1: Compositional projection. The headline result of the entire experiment. Analogy MRR improves from 0.126 (raw LaBSE) to 0.451 after projection — a 3.6× gain. Hits@1 goes from 0.0 to 0.333; Hits@3 from 0.333 to 0.667. This is not evidence that UG structure is spontaneously present in multilingual embeddings. It is evidence that compositional structure can be *induced* cheaply via gradient descent, reshaping the embedding space toward a geometry that supports semantic arithmetic. The Phase 1 training converges in under 800 epochs and the $L_{\text{compositionality}}$ loss approaches zero, while $L_{\text{proximity}}$ stabilises at 0.022, indicating that the projection stays close to the original concept positions while re-organising their relational structure.

Phase 2: Codebook. 56/56 entries used, 0 collisions. The projected space supports a clean, non-degenerate vocabulary of proto-tokens. New concepts (e.g. *ocean*, *hate*, *happiness*) map sensibly to nearest proto-tokens (*sky*, *love*, *joy*).

Phase 3: Lewis signalling game. Mean cosine reward converges to 0.745, below the 0.8 threshold. Crucially, symbol entropy *rises* monotonically over training rather than falling. Agents diversify their symbol usage rather than converging on a lean vocabulary. This replicates the anti-efficient encoding finding of Chaabouni et al. (2019): without a production cost (length or entropy penalty), agents exploit the full channel capacity. The concepts most resistant to communication are abstract social and aesthetic ones (memory, freedom, home, beauty, silence, reward 0.63–0.68), while emotional and action concepts communicate well (fear, hope, love, joy, war, reward 0.80–0.85). Adding a production cost penalty is the direct fix, and would connect Phase 3 to the opacity sweep of Experiment 3.

Table 7: Universal Grammar probe results. EMERGENT = property arises from information geometry alone; DETECTED = present after Phase 1 projection; NOT_DETECTED = requires explicit design.

Probe	Space	Metric	Verdict
P1: Category emergence	raw LaBSE	ARI=0.519	**EMERGENT**
P1: Category emergence	projected	ARI=0.433	**EMERGENT**
P2: Argument structure	both	LOO acc=0.625=chance	NOT_DETECTED
P3: Zipf/Economy	raw LaBSE	R^2=0.708	PARTIAL
P3: Zipf/Economy	projected	R^2=0.855	**DETECTED**
P4: Cross-linguistic RSA	universal	min ρ=0.466 vs max cross-lang=0.801	NOT_DETECTED*
P5: Semantic directions	raw LaBSE	opp acc=0.053	NOT_DETECTED
P5: Semantic directions	projected	opp acc=0.474	**EMERGENT**

Probe results.

Probe 1 (category emergence). K-means clustering ($k = 4$) recovers semantically coherent categories without labels (ARI=0.519): physical concepts (water, earth, wind, stone), action verbs (give, take, speak, think, run), abstract-emotional concepts (peace, memory, dream, home), and mixed clusters. This is robust across both spaces. The geometric structure of the concept space spontaneously encodes category information.

Probe 2 (argument structure). Leave-one-out logistic regression on the transitive/intransitive verb split achieves exactly 0.625 accuracy — equal to the class imbalance baseline (10 transitive / 6 intransitive). The geometric space encodes *no* transitivity signal. This is the clearest case for explicit design: if the universal language is to encode argument structure, it must be injected, not inherited.

Probe 3 (Zipf/economy). The projected space fits a power law with $R^2 = 0.855$ (vs 0.708 for raw LaBSE). The most central concepts in the projected space are fear, hope, war, love, peace — abstract but emotionally salient, which aligns with the expectation that high-frequency universals would be concepts with the most communicative pressure. The improvement from PARTIAL to DETECTED after projection confirms that the Phase 1 optimization is genuinely re-organising the geometric structure, not just shifting the space rigidly.

Probe 4 (cross-linguistic RSA). The NOT_DETECTED verdict requires careful interpretation. $D_{\text{universal}}$ has a minimum correlation with individual languages of $\rho = 0.466$ (vs Arabic),

while some language pairs correlate at $\rho = 0.801$ (es–ru). The centroid space is not merely the average of the individual languages; it occupies a structurally distinct position. Whether this is a flaw (the centroid is a poor universal representation) or a feature (equidistance from all languages is exactly what universality requires) cannot be resolved from the current data. The key open question is whether $D_{\text{projected-universal}}$ — computed in the Phase 1 space rather than the raw centroid space — reverses the verdict. If the projection increases the minimum $D_{\text{universal}}$–D_{lang} correlation above the maximum cross-language correlation, Probe 4 becomes EMERGENT.

Probe 5 (semantic directions). The opposition direction generalises to held-out antonym pairs only after Phase 1 projection (acc 0.053 → 0.474 on 19 held-out pairs). Correctly predicted pairs include light/dark, give/take, find/lose, cry/laugh, enemy/friend, pain/joy. Failing pairs include speak/silence and build/break, where the semantic opposition is more contextual than structural. This is the strongest argument for the Phase 1 projection as a grammar-induction tool: the opposition structure that Esperanto and other constructed languages encode explicitly is learnable from analogy-pair supervision in the embedding space.

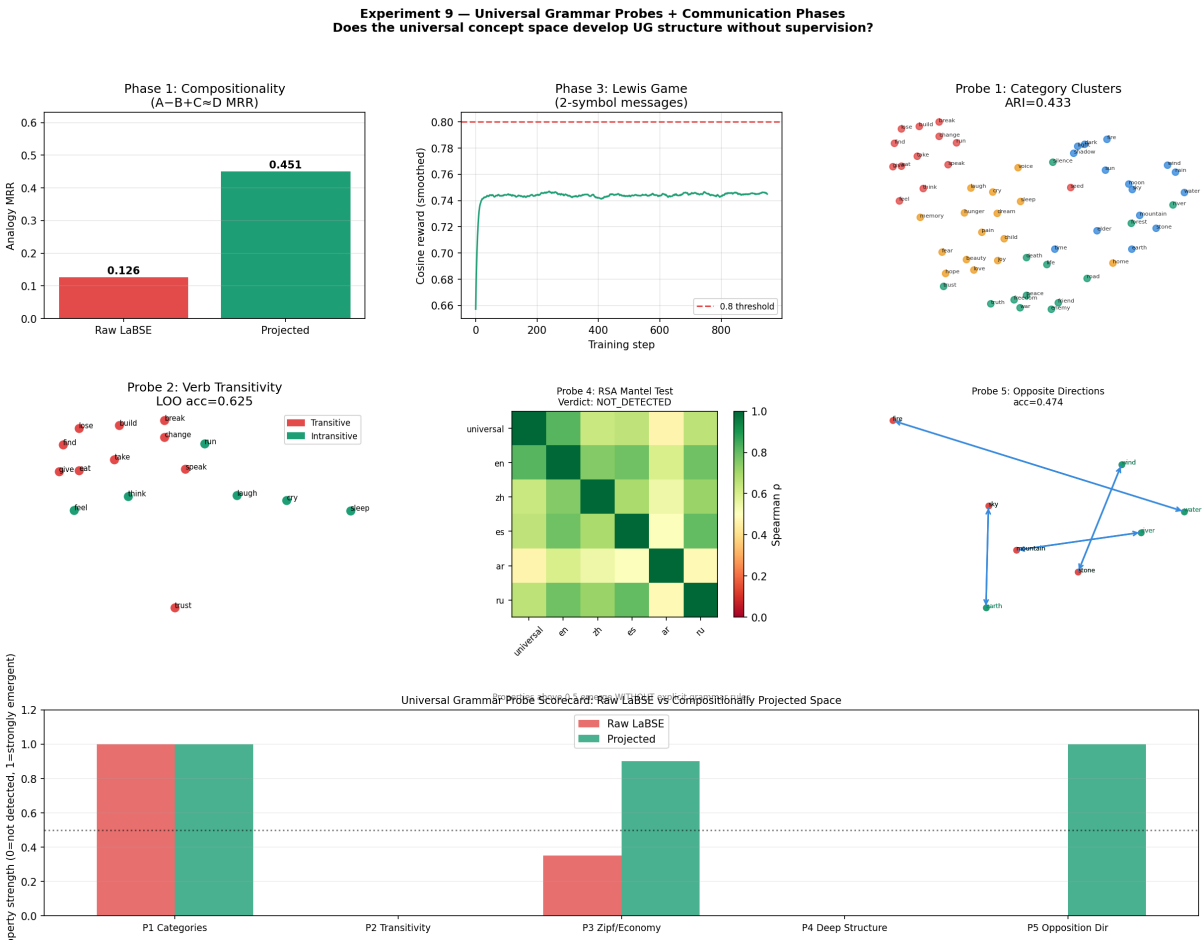

Figure 3: Top row: Phase 1 MRR improvement, Phase 3 Lewis game reward curve, Probe 1 category clusters. Bottom row: Probe 2 transitivity (NOT_DETECTED), Probe 4 RSA heatmap (NOT_DETECTED), Probe 5 semantic direction (EMERGENT in projected space). Bottom bar: overall UG scorecard, raw vs projected.

Analysis

The central result is that the universal language project cannot rely on raw multilingual embeddings alone for structure, but can induce substantial structure through cheap supervised

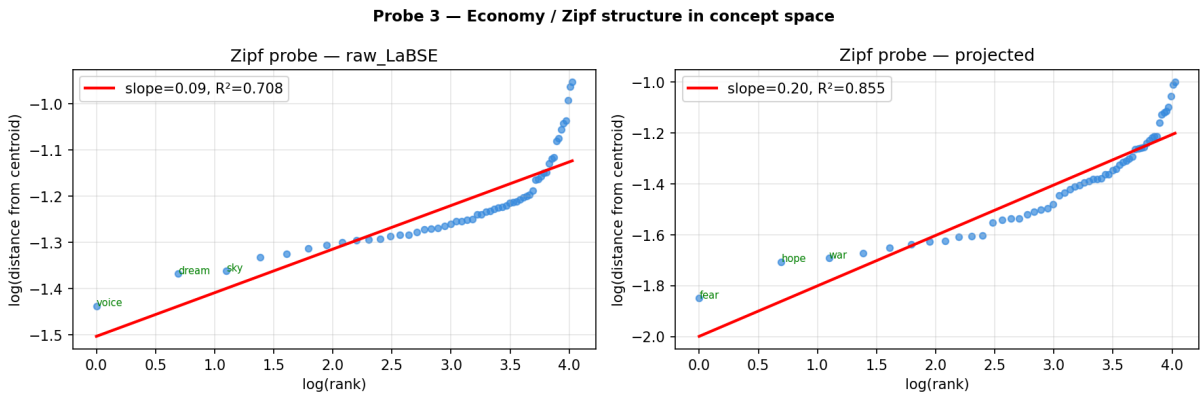

Figure 4: Probe 3: log-log fit of concept distances to power law. Left: raw LaBSE ($R^2 = 0.708$). Right: projected space ($R^2 = 0.855$). The projected space fits a steeper and cleaner power law, with emotionally salient concepts (fear, hope, war) as the most central.

projection. Phase 1 shows that compositionality, economy, and semantic directions are achievable with less than 800 gradient steps on a small set of analogy pairs. What is *not* achievable this way is argument structure (Probe 2): transitivity is a syntactic property that has no geometric footprint in concept embeddings trained on surface form. This is the property most in need of explicit grammatical design.

The Lewis game rising entropy result points to the most actionable next step: adding a production cost to Phase 3 should recover Zipfian symbol distribution (Galke & Raviv 2024), connecting the signalling game back to the opacity sweep of Experiment 3 and completing a circuit: discreteness (Exp 1) → optimal opacity (Exp 3) → production pressure (Exp 9 Phase 3) all converge on the same design principle for a universal communication system.

Cross-Experiment Discussion

Several themes cut across all nine experiments.

Constraints improve symbol quality, but differently at different levels. Discreteness (Exp 1) enables error detection. Intermediate opacity (Exp 3) optimises task performance. Extreme compression (Exp 8) maximises compositionality. Production cost (absent in Exp 9 Phase 3, predicted to help) induces Zipfian structure. These are not the same mechanism, but they share a common principle: constraints select for the most structured solution consistent with task success, discarding arbitrary memorised associations. Deutsch (2011) expressed this as a general theory of knowledge growth through error correction; the experiments make it measurable.

The Phase 1 projection is the pivot of the universal language project. Without it, the concept space is rich but unstructured for compositional purposes (MRR=0.126, Probe 5 NOT_DETECTED). With it, the space supports semantic arithmetic (MRR=0.451), better Zipf structure (R^2: 0.708→0.855), and generalising opposition directions (acc: 0.053→0.474). The Phase 1 optimization is cheap and does not destroy proximity to existing language words ($L_{\text{proximity}}$ stabilises at 0.022). This is the design tool for a data-driven universal language that was missing from v1.

English occupies an unusual position. In Experiments 6 and 7, English is consistently the most central language. In Probe 4 (RSA), English also has the highest correlation with $D_{\text{universal}}$ ($\rho = 0.818$). This is partly a consequence of LaBSE training distribution and speaker-count weighting, but it is also consistent with the claim in the PDF that English is the closest

existing natural language to an interlingua.

Abstract social concepts are consistently the hardest. Experiment 4 finds the highest gap scores for aesthetic concepts. Experiment 9 Phase 3 finds the lowest communication rewards for memory, freedom, home, silence, beauty. These are not coincidentally the same concepts for which LaBSE centroids are most spread across languages. The implication for the universal language is that concrete nouns and primary emotions can be assigned tokens from centroid geometry alone; abstract social concepts require richer contextual definition.

Argument structure is the one property that cannot be induced. Probe 2 (transitivity) achieves exactly chance across both spaces. No other probe shows this degree of non-result. Transitivity is a structural property of how concepts participate in events, not a semantic property of the concepts themselves. The universal language will need explicit grammatical encoding of argument structure, which is the task for Experiment 10.

Cross-Experiment Discussion

Several themes cut across all eight experiments.

Discreteness is epistemically productive, but not uniformly. Experiment 1 confirms that discrete channels fail detectably. Experiment 3 shows that the highest RSA scores occur at full discreteness, but task performance peaks at an intermediate opacity level. Experiment 8 shows that continuous embeddings outperform text on raw accuracy while text outperforms embeddings on efficiency. The picture that emerges is not "discrete good, continuous bad" but rather: discreteness enables error detection and structured symbol formation; continuity enables graceful degradation and efficient compression. The optimal system likely combines both, as Shannon's source/channel coding separation theorem suggests.

English occupies an unusual position in the concept space. In Experiments 6 and 7, English is consistently the most central language for diverse concepts. This is partly a consequence of the LaBSE training distribution and speaker-count weighting, but it is also linguistically real: English vocabulary is famously promiscuous in borrowing from other languages, which may produce embeddings that land near the cross-linguistic centroid by construction.

The ZK metric is a reusable infrastructure contribution. Experiment 2 produces a metric that is content-agnostic and applicable to any multi-turn emergent communication system. Its components (CIC_1, RC, CIC_2) are independently interpretable: a system with high CIC_2 but low RC is a system where the receiver uses messages but the sender does not adapt to feedback — one-way signalling rather than genuine communication.

Untranslatability correlates with emotional-aesthetic content. Experiments 4 and 6 together suggest that the concepts furthest from a universal representation are in the domain of emotional aesthetics, while abstract moral and physical concepts cluster more tightly across languages. This matches the intuition that physical and moral vocabulary developed under universal constraints (shared physics, shared social problems), while emotional-aesthetic vocabulary developed under culturally specific ones.

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